

Taylor Kirk

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SKILLS

Generative AI & LLMs : LangChain, LangGraph, RAG, Fine-tuning, Prompt Engineering, OpenAI API, Anthropic
Data Analysis & ML : Python, R, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn
Infrastructure & Tools : Postgres, Redis, MLFlow, Hugging Face, Azure, AWS Sagemaker, Streamlit, Git

EXPERIENCE

Sweeten Health Jan. 2026 – Present
Data Analyst Remote

- Automated KPI and goal-tracking pipelines using **n8n** and **JS**, eliminating ~8–10 hours of manual weekly reporting.
- Built static and interactive dashboards (**Plotly Dash**, HTML/CSS) for internal analysis; two presented in client-facing meetings.
- Integrated patient data across **EMR systems**, CRM portals, and billing sources to support CCM program analytics and revenue reporting.

LiaZen Mar. 2025 – Oct. 2025
AI Engineering Intern San Diego, CA

- Architected a multi-node **LangGraph** workflow for real-time communication moderation, utilizing **Redis** for stateful session management and **Postgres** for persistent audit logging.
- Integrated **LangChain** guardrails to detect and reshape toxic communication while maintaining conversational tone.

Upwork (Freelance Contract) Jan. 2025 – Sept. 2025
AI/ML Engineer Remote

- Engineered a therapy-aid application using **LLMs** to assist students through role-play of specific therapeutic techniques.
- Developed **ML pipelines** with **scikit-learn** and **MLflow** to classify and evaluate patient-therapist interactions.
- Designed a **synthetic data generation** workflow to fine-tune models for realistic and diverse patient profiles.

Aya Healthcare Oct. 2014 – Sept. 2024
Senior Recruiter San Diego, CA

- Placed **1,500+ travel clinicians**, contributing to organizational growth from 600 to 40,000+ employees.

TECHNICAL PROJECTS

Offline RL for Clinical Decision Support | *Python, PyTorch, LSTM, CQL, MIMIC-IV* | *Live App*

- Built a two-stage pipeline on ~396K ED stays — **LSTM** sequence model (F1 0.93, PR-AUC 0.98) feeding a **CQL** offline RL agent that achieved **F1 macro 0.85** and flagged ICU transfer a median of **4 clinical events earlier** than the treating provider.

Sentiment Analysis for Political Discourse | *Python, Transformers, BERTweet, Hugging Face*

- Fine-tuned **BERTweet-large** on political social media comments for discourse classification across a custom 5-label schema (Argumentative, Informational, Opinion, Expressive, Neutral), achieving **78.17% F1 macro**; dataset and model published to **Hugging Face**.

EDUCATION

University of San Diego May 2024 – Apr. 2026
M.S. Applied Data Science San Diego, CA

- **Coursework**: Applied Predictive Modeling, Time Series Analysis, Machine Learning, Applied LLMs.

CERTIFICATIONS

AI Essentials | *Ready Tensor (RAG Expert)*